

Scientific Computation

Federico Formica
Material by: Prof. Matthew Giamou
Nov. 6, 2024

Lecture 23 - Interpolation

Summary

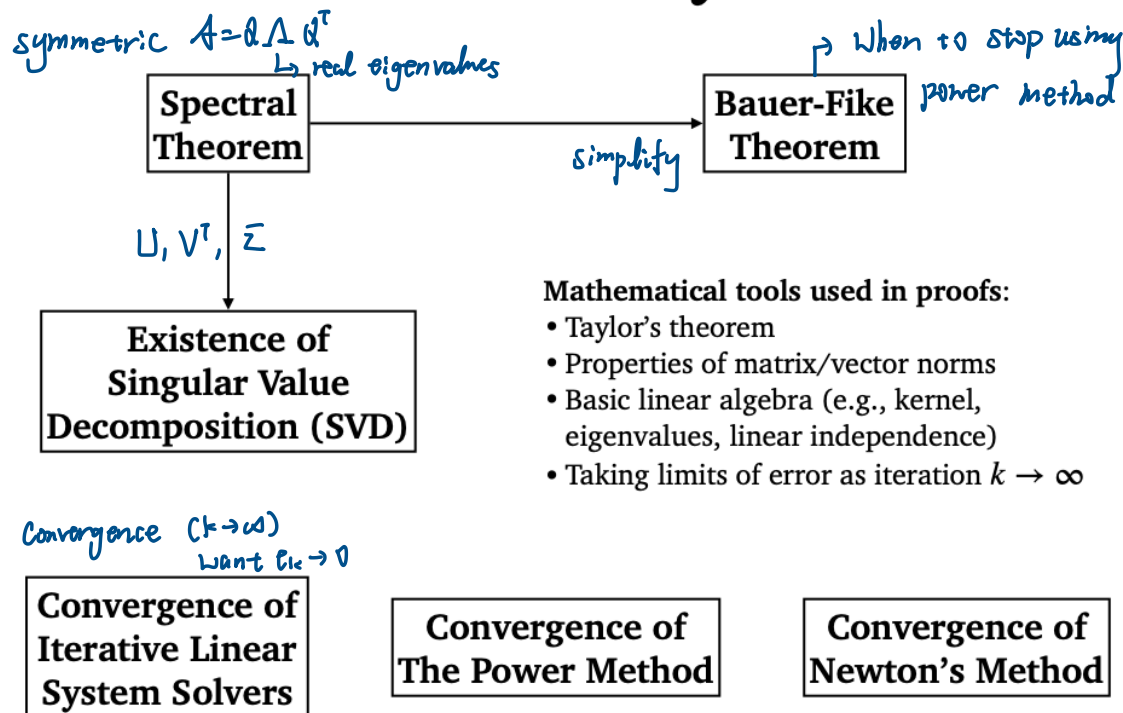
- Floating point numbers introduce roundoff error
 - We can predict bounds on this error
- Roundoff error affects solution of linear systems
 - Pivoting partially mitigates this error
 - Some problems $Ax = b$ are harder than others (e.g., condition number of A matters)
 - We can exploit structure in linear systems (e.g., symmetry ($A = A^T$), positive definiteness ($A > 0$), sparsity (iterative methods)).
- Eigenvalues/eigenvectors and the singular value decomposition (SVD) help us characterize matrices
- Taylor series helps us to linearize various problems, allowing us to

$$\begin{aligned} f(x) &= x(1+\delta) \\ |\delta| &\leq u \end{aligned}$$

- Taylor series helps us to linearize various problems, allowing us to apply linear tools
 - Root finding, optimization, **interpolation**, integration, ODEs...

2

Theorems Recently Covered



3

Lecture Outline

- Example of multivariate optimization
- Interpolation
- Representation and basis functions

- Monomial basis functions

A Simple Optimization Example

Example 2: Consider the nonlinear unconstrained optimization problem

$$\min_{x \in \mathbb{R}^2} f(x)$$

where $f(x) = \frac{1}{4}(x_1 - 1)^4 + (x_1x_2 - \frac{1}{2})^2 + 1.0$.

5

Newton's Method for Example 2

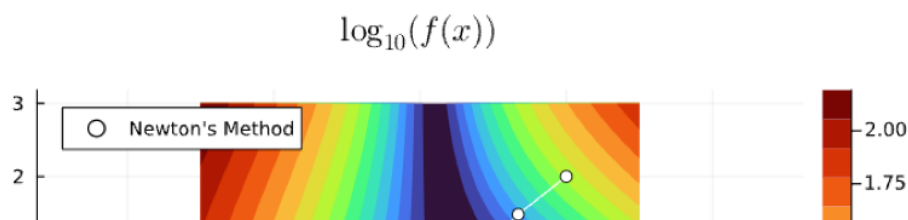
Example 2 cont.: The Hessian of this function is:

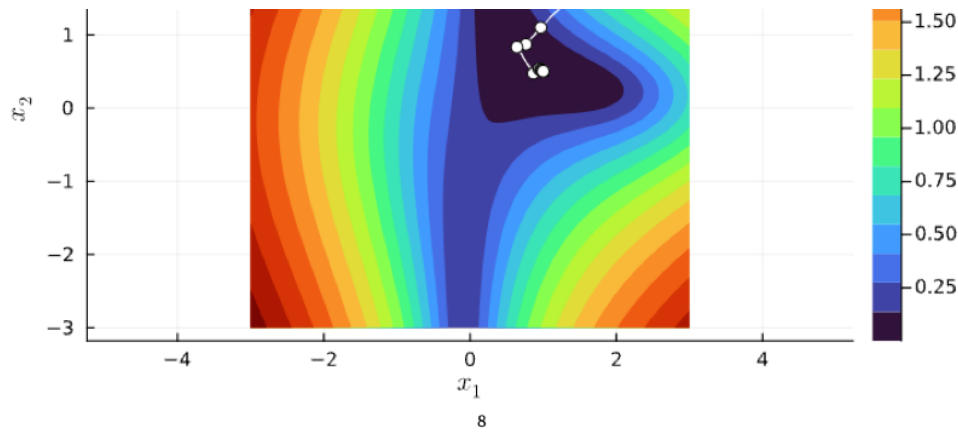
Solving Example 2 by Hand

Example 2 cont.: We can find the solution for this toy problem by inspection and check the sufficient optimality conditions:

Example 2 Visualized

Here is a plot of Newton's method initialized with $x^{(0)} = [2 \ 2]^T$ (note that the cost function is passed through $\log_{10}$ for effect):





Lecture Outline

- Example of multivariate optimization
- **Interpolation**
- Representation and basis functions
- Monomial basis functions

The Interpolation Problem

n + 1 points

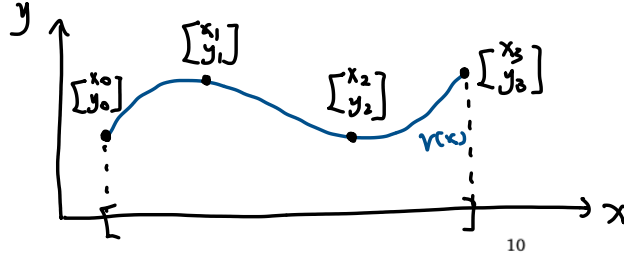
Given data points $\{(x_i, y_i)\}_{i=0}^n$, find a function $v(x)$ such that

$$v(x_i) = y_i, \quad i = 0, \dots, n,$$

i.e., the **interpolant** $v(x)$ **exactly fits** the data points. We are only interested in querying at

$$x \in [\min(\{x_i\}_{i=0}^n), \max(\{x_i\}_{i=0}^n)],$$

i.e., we are not **extrapolating** beyond the domain of our dataset.



10

Applications

→ don't have access

Approximating functions: if $f(x)$ is expensive to compute/collect, we can find a simpler $v(x)$ that approximates it for $x \neq x_0, x_1, \dots, x_n$

The interpolant v may also be easier to differentiate or integrate than f → have access but complex
(e.g., it is smoother, or has an analytical derivative/antiderivative)

→ Machine learning (different setup than regression)

→ Geophysical / environmental data

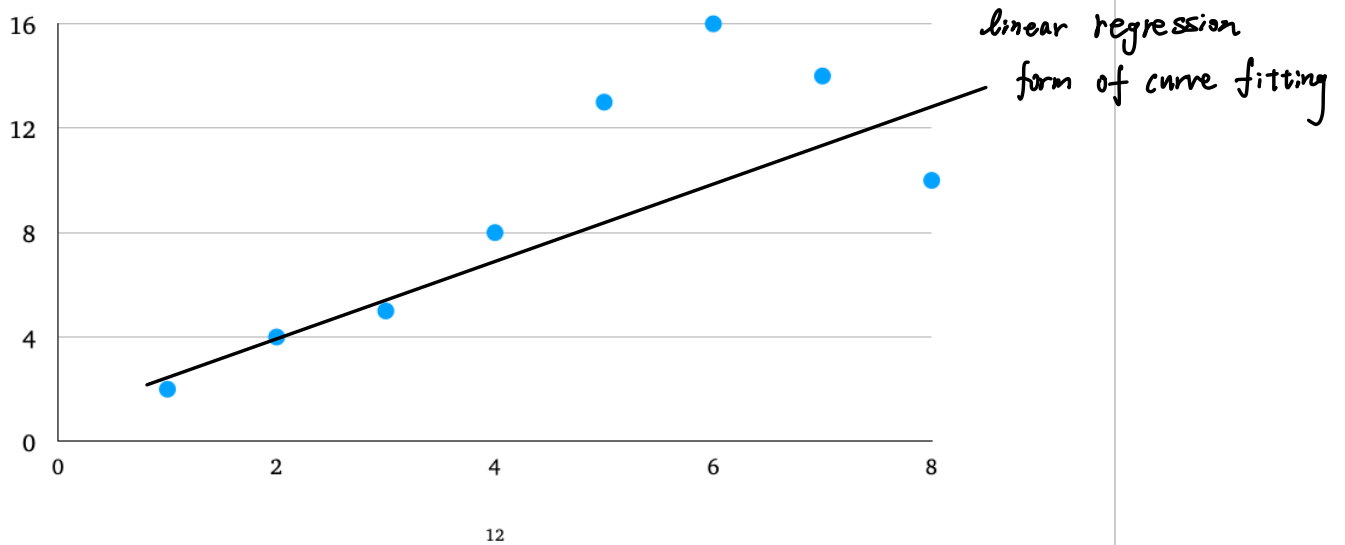
→ Smooth motion planning / state estimation

→ Graphics and animation

11

Interpolation vs. Curve Fitting

We saw that **curve fitting** or **smoothing** is a related problem which removes the requirement that $v(x_i) = y_i$ for all data points $i = 0, \dots, n$.



Lecture Outline

- Example of multivariate optimization
- Interpolation
- **Representation and basis functions**
- Monomial basis functions

Representation

Let us assume that our interpolant v takes the form

Let us assume that our interpolant v takes the form

$$v(x) = \sum_{j=0}^n c_j \phi_j(x) = c_0 \phi_0(x) + c_1 \phi_1(x) + \dots + c_n \phi_n(x),$$

where c_j are unknown coefficients and ϕ_j are given **basis functions**.

↓
These are what we solve
for as a function of the
observed dataset $\{(x_i, y_i)\}_{i=0}^n$

↳ seen this in regression
↳ These define the interpolation
algorithm we are using

Why basis function?

14

$V(x) = 0$

Linear Independence

We want our basis functions $\phi(x)$ to be **linearly independent**: → similar sense as vectors in $\mathbb{R}^n$

$$v(x) = \sum_{j=0}^n c_j \phi_j(x) = 0 \quad \forall x \implies c_j = 0 \quad \forall j,$$

i.e., we treat the functions $\phi(x)$ like **vectors**. Recall from linear algebra:

Basis vectors $\{b_0, b_1, \dots, b_n\}$ lin. indep $\iff$ each b_i cannot be written as

$\implies$ There is no $C \in \mathbb{R}^{n+1}$ s.t. $C \neq 0$ and

$$B_c = [b_0, b_1, \dots, b_n] \quad C = \sum_{i=0}^n c_i b_i = 0$$

↳ i.e. B is full rank

a linear combination
of the others.

15

Linear System

Let's look at the actual equations for our dataset $\{(x_i, y_i)\}_{i=0}^n$:

Square matrix X

$$\begin{bmatrix} \phi_0(x_0) & \phi_1(x_0) & \cdots & \phi_n(x_0) \\ \phi_0(x_1) & \phi_1(x_1) & \cdots & \phi_n(x_1) \\ \vdots & \vdots & & \vdots \\ \phi_0(x_n) & \phi_1(x_n) & \cdots & \phi_n(x_n) \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} y_0 \\ y_1 \\ \vdots \\ y_n \end{bmatrix},$$

where our goal is to solve for $c \in \mathbb{R}^{n+1}$.

X is full rank (col/row are lin. indep.) $\Leftrightarrow$ unique solution $C = X^T y$

How do we ensure whether X is full rank?

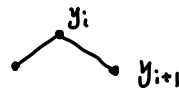
If we choose lin. indep. basis function ϕ_j , the columns of X are lin. indep.
for any unique x :

16

Choice of Basis Functions

In this framework, designing an interpolation algorithm boils down to selecting a linearly independent set of basis functions. Options include:

Piecewise interpolation, e.g.:



$$\phi_j(x) = \begin{cases} y_i + (x - x_i) \frac{y_{i+1} - y_i}{x_{i+1} - x_i}, & \text{if } x \in [x_i, x_{i+1}] \\ 0, & \text{otherwise.} \end{cases}$$

show that piecewise lin. functions are lin. indep.

Trigonometric functions (think Fourier series), e.g.:

$$\phi_j(x) = \cos(jx), j = 0, 1, \dots, n$$

$\hookrightarrow$ data is periodic (i.e. comms)

Monomials, e.g.:

$\hookrightarrow$ we'll focus on
in this course

$$\phi_j(x) = x^j, j = 0, 1, \dots, n$$

$\hookrightarrow$ gives you the family of n^{th} degree polynomials

17

Lecture Outline

- Example of multivariate optimization
- Interpolation
- Representation and basis functions
- **Monomial basis functions**

Monomial Interpolation

We will consider simple **monomials** $\rightarrow$ *polynomial with one term*

$$\phi_j(x) = x^j, j = 0, 1, \dots, n$$

$\hookrightarrow$ *basis vectors*

for our interpolant

$$\begin{aligned} v(x) &= \sum_{j=0}^n c_j \phi_j(x) = c_0 \phi_0(x) + c_1 \phi_1(x) + \dots + c_n \phi_n(x), \\ &= \sum_{j=0}^n c_j x^j \end{aligned}$$

where

$$v(x_i) = y_i, i = 0, \dots, n$$

for our dataset $\{(x_i, y_i)\}_{i=0}^n$.

Simple Example

Example 1: Interpolate the dataset

$$\begin{array}{c} 0 \quad 1 \quad 2 \quad (n=2) \\ x_i: \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} \\ y_i: \begin{bmatrix} 1 \\ 3 \\ 3 \end{bmatrix} \end{array}$$

using a polynomial of degree 2. We seek the coefficients of

$$p_2(x) = c_0 + c_1x + c_2x^2. \quad \text{space of quadratic polynomials}$$

solve with $p_2(x_i) = y_i$

$$p_2(1) = c_0 + c_1 + c_2 = 1$$

$$p_2(2) = c_0 + 2c_1 + 4c_2 = 3$$

$$p_2(4) = c_0 + 4c_1 + 16c_2 = 3$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 4 & 16 \end{bmatrix} c = \begin{bmatrix} 1 \\ 3 \\ 3 \end{bmatrix}$$

20

Simple Example

Example 1 cont.: This amounts to solving the linear system

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 4 & 16 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 3 \end{bmatrix}$$

```
using LinearAlgebra
```

```
x_data = [1; 2; 4]
```

```
y_data = [1; 3; 3]
```

```
A = [ones(size(x_data, 1)) x_data x_data.^2]
```

```
display(A)
```

```
c = A \ y_data
```

```
3x3 Matrix{Float64}:
```

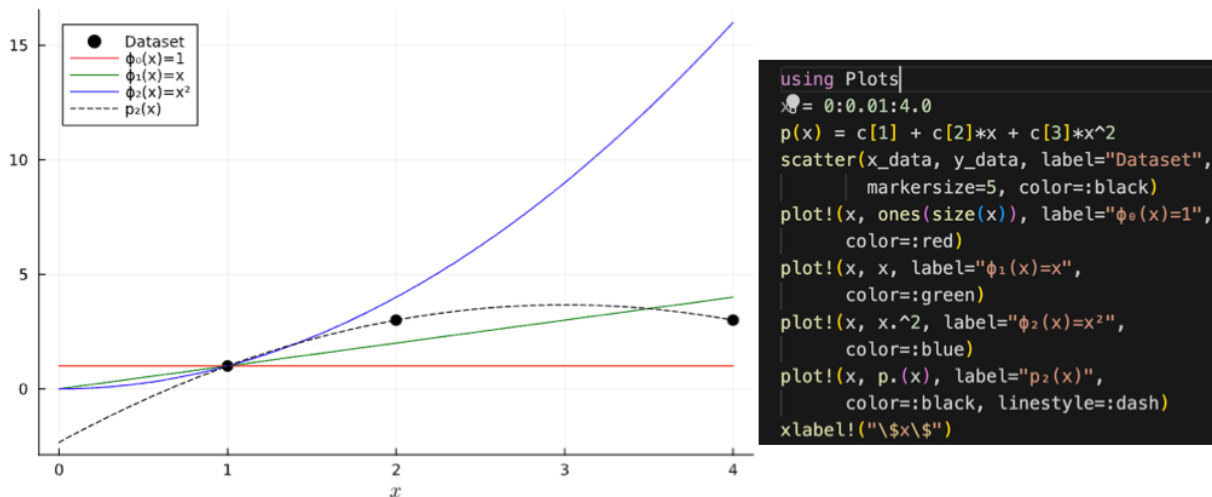
```
1.0  1.0  1.0
1.0  2.0  4.0
1.0  4.0  16.0
```

```
3-element Vector{Float64}:
```

```
-2.3333333333333334
 4.0000000000000001
-0.6666666666666669
```

$$p_2(x) = -\frac{7}{3} + 4x - \frac{2}{3}x^2$$

Example 1 Visualized



Vandermonde Matrix

In general, we have the linear system

$$\sum_{i=1}^n a_i Xc = \begin{bmatrix} 1 & x_0 & x_0^2 & \cdots & x_0^n \\ 1 & x_1 & x_1^2 & \cdots & x_1^n \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^n \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} y_0 \\ y_1 \\ \vdots \\ y_n \end{bmatrix},$$

where the coefficient matrix X is called a **Vandermonde matrix**.

$$\det(X) = \prod_{i=0}^{n-1} \prod_{j=i+1}^n (x_j - x_i)$$

When $x_i = x_j$ for some $i \neq j$, $\det(X) = 0$

product notation

$$\prod_{i=1}^n a_i = a_1 a_2 \cdots a_n$$

$$\prod_{i=1}^n a_i = a_1 \cdot a_2 \cdots a_n$$

23

Uniqueness of the Interpolant

Therefore, if the x_i are all unique, then the determinant

$$\det(X) = \prod_{i=0}^n \left[\prod_{j=i+1}^n (x_j - x_i) \right] \neq 0,$$

and we know that:

$\det(X) \neq 0 \Rightarrow X$ is non singular (invertible)

$\Rightarrow$ the system has a unique solution

$\Rightarrow$ there is a unique polynomial of degree n or less that interpolates our $n+1$ points

However: work to solve for C is $O(n^3)$

difficult to add new points x_{n+1}, y_{n+1} ($\text{cond}(A) \gg 1$)
 X can have a large condition number

24

Interpolation 和 linear regression 的区别是, linear regression 是尽量接近数据点, interpolation 是要求一定要穿过所有数据点。我们只在给定的数据点范围内使用这个插值函数, 不会考虑范围之外的点。

插值问题要求使用的 basis function 必须是 linear independent 的, 这样保证的插值解的唯一性; basis function 可以选很多种, 比如说 monomial polynomial, piecewise 或者 Fourier。

Vandermonde 矩阵证明了插值的唯一性, 但是由于其大部分时间 condition number 都很大, 是 ill-conditioned, 所以很难算。